

Chapter 7

Representation and Eigenvalues

This chapter wants to bound the eigenvalues of Cayley graphs.

7.1 Decompose Adjacency Operator

Consider a Cayley graph $\text{Cay}(G, \Gamma)$ with adjacency operator A , let $f \in L^2(G)$, and let R be the right regular representation of G . Then

$$(Af)(x) = \sum_{\gamma \in \Gamma} f(x\gamma) = \sum_{\gamma \in \Gamma} (R(\gamma)f)(x)$$

So we have $A = \sum_{\gamma \in \Gamma} R(\gamma)$

Proposition 7.1. Finite group G , $\Gamma \subseteq G$, adjacency operator A of $\text{Cay}(G, \Gamma)$. If $\pi_1, \dots, \pi_k$ is a complete set of inequivalent matrix irreps of G , then

$$A \cong d_1 M_{\pi_1} \oplus \dots \oplus d_k M_{\pi_k}$$

where $d_i = \dim(\pi_i)$, $M_\pi = \sum_{\gamma \in \Gamma} \pi(\gamma)$

Proof

By Theorem 6.30 and the fact above,

$$A = \sum_{\gamma \in \Gamma} R(\gamma) \cong \sum_{\gamma \in \Gamma} (d_1 \pi_1(\gamma) \oplus \dots \oplus d_k \pi_k(\gamma))$$

Corollary 7.2. Group G , $\Gamma, \Gamma' \subseteq G$ s.t. $\Gamma \subset \Gamma'$. Let $X = \text{Cay}(G, \Gamma)$, $X' = \text{Cay}(G, \Gamma')$. Then $|\Gamma| - \lambda_1(X) \leq |\Gamma'| - \lambda_1(X')$.

Remark

That is, increasing the size of Γ would never decrease the spectral gap.

Proof

Let A, A' be adjacency operators for X, X' . If $f \in L^2(G)$ with $\|f\|_2 = 1$, then by Cauchy and the fact that R is unitary w.r.t. the standard inner product,

$$|\langle R(\gamma)f, f \rangle_2| \leq \|R(\gamma)f\|_2 \|f\|_2 = \|f\|_2^2 = 1$$

for all $\gamma \in \Gamma$.

Suppose $\gamma = \gamma^{-1}$,

$$\langle R(\gamma)f, f \rangle_2 = \langle R(\gamma^{-1})R(\gamma)f, R(\gamma^{-1})f \rangle_2 = \langle f, R(\gamma)f \rangle_2 = \overline{\langle R(\gamma)f, f \rangle_2}$$

so $\langle R(\gamma)f, f \rangle_2$ is real.

For any γ ,

$$\langle R(\gamma)f, f \rangle_2 + \langle R(\gamma^{-1})f, f \rangle_2 = \langle f, R(\gamma^{-1})f \rangle_2 + \langle f, R(\gamma)f \rangle_2 = \overline{\langle R(\gamma)f, f \rangle_2} + \langle R(\gamma^{-1})f, f \rangle_2$$

so $\langle R(\gamma)f, f \rangle_2 + \langle R(\gamma^{-1})f, f \rangle_2$ is real.

Let f as defined above,

$$\begin{aligned} |\Gamma| - \langle Af, f \rangle_2 &= |\Gamma| - \sum_{\gamma \in \Gamma} \langle R(\gamma)f, f \rangle_2 = \sum_{\gamma \in \Gamma} (1 - \langle R(\gamma)f, f \rangle_2) \\ &\leq \sum_{\gamma' \in \Gamma'} (1 - \langle R(\gamma')f, f \rangle_2) = |\Gamma'| - \sum_{\gamma' \in \Gamma'} \langle R(\gamma')f, f \rangle_2 \\ &= |\Gamma'| - \langle A'f, f \rangle_2 \end{aligned}$$

The inequality relies on $|\langle R(\gamma)f, f \rangle_2| \leq 1$, and each new element of Γ' would contribute a real number in the summation (either one to one, or a pair of elements contributing one). Then, by Rayleigh Ritz, choose f s.t. $\lambda_1(X') = \langle A'f, f \rangle$ and $\|f\|_2 = 1$

$$|\Gamma| - \lambda_1(X) \leq |\Gamma| - \langle Af, f \rangle \leq |\Gamma'| - \langle A'f, f \rangle = |\Gamma'| - \lambda_1(X')$$

Note two graph share the same vertex set, thus same $L^2(G)$.

7.2 Unions of Conjugacy Classes

This section discusses how to find the eigenvalues when Γ is a union of conjugacy classes.

Proposition 7.3. Finite group G , $\Gamma \subseteq G$ s.t. $g\Gamma g^{-1} = \Gamma$ for all $g \in G$. Let $X = \text{Cay}(G, \Gamma)$, adjacency operator A . Let $\rho_1, \dots, \rho_r$ be a complete set of inequivalent irreps of G , let χ_i be the character of ρ_i , d_i the degree of ρ_i . Then the eigenvalues of A are given by

$$\mu_i = \frac{1}{d_i} \sum_{\gamma \in \Gamma} \chi_i(\gamma)$$

for $i = 1, \dots, r$, where each eigenvalue μ_i occurs with multiplicity d_i^2

Remark

It is possible that $\mu_i = \mu_j$ for $i \neq j$. Therefore, we should combine multiplicities when two μ overlaps. That is, μ_i occurs with multiplicity $\sum d_j^2$, where j is s.t. $\mu_i = \mu_j$

Proof

By Theorem 6.30, we can decompose $L^2(G) = \bigoplus_{i=1}^r d_i V_i$ where $V_1, \dots, V_r$ corresponds to a complete list of inequivalent irreps of G , and $R(\gamma)v_i = \rho_i(\gamma)v_i$ for all $v_i \in V_i$ (invariant subspace). Therefore, the restriction of $R(\gamma)$ to V_i is $\rho_i(\gamma)$

Since Γ is closed under conjugation, we have

$$AR(g) = \sum_{\gamma \in \Gamma} R(\gamma)R(g) = \sum_{\gamma \in \Gamma} R(\gamma g) = \sum_{\gamma \in \Gamma} R((g\gamma g^{-1})g) = \sum_{\gamma \in \Gamma} R(g\gamma) = R(g)A$$

for all $g \in G$

Fix i , consider the restriction of R and A to V_i . Since $A = \sum_{\gamma \in \Gamma} R(\gamma)$ and $R : V_i \rightarrow V_i$, we have $A : V_i \rightarrow V_i$. Pick basis β of V_i , then $[R(g)]_\beta [A]_\beta = [A]_\beta [R(g)]_\beta$ for all $g \in G$. Since R restricted to V_i is the irreducible representation ρ_i , and we are dealing with matrix representations, by Corollary 6.23, $[A]_\beta = \mu_i I$, where I is the $d_i \times d_i$ identity matrix and $\mu_i \in \mathbb{C}$. Consider the trace of the matrix representation of A restricted to V_i w.r.t. basis β , denote this thing as $[A|_{V_i}]_\beta$, we have

$$d_i \mu_i = \text{tr}([A|_{V_i}]_\beta) = \text{tr}\left(\sum_{\gamma \in \Gamma} [\rho_i(\gamma)]_\beta\right) = \sum_{\gamma \in \Gamma} \chi_i(\gamma)$$

Thus, $\mu_i = \frac{1}{d_i} \sum_{\gamma \in \Gamma} \chi_i(\gamma)$

Repeat the process for each i , and choose bases for each V_i carefully s.t. they can form a basis for $L^2(G)$, then we can see

$$A \cong d_1(\mu_1 I_{d_1}) \bigoplus \dots \bigoplus d_r(\mu_r I_{d_r})$$

Recall the notation, where $d_1(\mu_1 I_{d_1})$ means d_1 copies of $(\mu_1 I_{d_1})$ direct summed together. Therefore, the eigenvalues of A are $\mu_1, \dots, \mu_r$, where μ_i has multiplicity d_i^2 .

Remark

Group G , $\Gamma \subseteq G$ s.t. $\Gamma = \bigcup_{i=1}^n K_{g_i}$ for $g_1, \dots, g_n \in G$, where K_{g_i} is the conjugacy class of g_i . Since characters are constant on conjugacy classes,

$$\mu_i = \frac{1}{d_i} \sum_{\gamma \in \Gamma} \chi_i(\gamma) = \frac{1}{d_i} \sum_{j=1}^n \sum_{h \in K_{g_j}} \chi_i(h) = \frac{1}{d_i} \sum_{j=1}^n |K_{g_j}| \chi_i(g_j)$$

7.3 An Upper Bound on $\lambda(X)$

Proposition 7.4. Finite group G , $\Gamma \subseteq G$, $X = \text{Cay}(G, \Gamma)$, assume X is non-bipartite. Let $\chi_1, \dots, \chi_r$ be a complete list of irreducible, inequivalent, nontrivial characters of G . Let d_i be the degree of χ_i . For any positive integer m , we have

$$\lambda(X) \leq \left(\sum_{k=1}^r d_k \sum_{g \in G} N_g \chi_k(g) \right)$$

where N_g is the number of ways to express g as a product of $2m$ elements (not necessarily distinct) elements of Γ .

Remark

$\lambda(X)$ here is any eigenvalue of X .

Proof

Let $d = |\Gamma|$, $n = |X|$, π_i be irreducible matrix representation corresponding to χ_i . By Prop. 7.1, and since the right regular representation collects all irreducible representations,

$$A \cong (d) \bigoplus d_1 M_{\pi_1} \bigoplus \dots \bigoplus d_r M_{\pi_r}$$

where $M_{\pi_i} = \sum_{\gamma \in \Gamma} \pi_i(\gamma)$. The first component corresponds to "the" trivial representation (which is irreducible and of degree 1), is a 1×1 matrix with entry d . The remaining eigenvalues lie in these nontrivial irreps. Since X is non-bipartite, $-d$ does not occur as an eigenvalue.

Let

$$\hat{A} = d_1 M_{\pi_1} \bigoplus \dots \bigoplus d_r M_{\pi_r}$$

be the nontrivial part of A

7.4 Eigenvalues: Dihedral Groups

Here we discuss the case when n is even for dihedral group D_n . Recall the dihedral group is given by

$$D_n = \{1, r, \dots, r^{n-1}, s, sr, \dots, sr^{n-1}\}$$

where $r^n = 1, s^2 = 1, rs = sr^{-1}$, which implies $r^i s = sr^{-i}$.

Proposition 7.5. If n is even, then a complete list of inequivalent, irreducible matrix representations of the dihedral group D_n is given as follows.

- 4 matrix representations of degree 1

	r^k	sr^k
ψ_1	1	1
ψ_2	1	-1
ψ_3	$(-1)^k$	$(-1)^k$
ψ_4	$(-1)^k$	$(-1)^{k+1}$

- $\frac{n}{2} - 1$ matrix representations of degree 2

$$\pi_h(r^k) = \begin{pmatrix} \xi^{hk} & 0 \\ 0 & \xi^{-hk} \end{pmatrix}, \quad \pi_h(sr^k) = \begin{pmatrix} 0 & \xi^{-hk} \\ \xi^{hk} & 0 \end{pmatrix}$$

where h is an integer with $1 \leq h \leq \frac{n}{2} - 1$ and $\xi = e^{2\pi i/n}$

Proof

It is straightforward to show that these are indeed homomorphisms (hence matrix representations). Also, it is clear that one-dimensional representations are irreducible and inequivalent.

Let χ_h be the character of π_h . Suppose $1 \leq h_1 \leq h_2 \leq \frac{n}{2} - 1$, then

$$\begin{aligned} \langle \chi_{h_1}, \chi_{h_2} \rangle_2 &= \sum_{k=0}^{n-1} \chi_{h_1}(r^k) \overline{\chi_{h_2}(r^k)} + \sum_{k=0}^{n-1} \chi_{h_1}(sr^k) \overline{\chi_{h_2}(sr^k)} \\ &= \sum_{k=0}^{n-1} (\xi^{h_1 k} + \xi^{-h_1 k})(\xi^{-h_2 k} + \xi^{h_2 k}) \\ &= \sum_{k=0}^{n-1} (\xi^{(h_1-h_2)k} + \xi^{(h_1+h_2)k} + \xi^{-(h_1+h_2)k} + \xi^{(h_2-h_1)k}) \end{aligned}$$

Since $1 \leq h_1 \leq h_2 \leq \frac{n}{2} - 1$, $0 < h_1 + h_2 < n$. Thus, $\xi^{h_1+h_2} \neq 1$, then

$$\sum_{k=0}^{n-1} \xi^{(h_1+h_2)k} = \frac{\xi^{(h_1+h_2)kn} - 1}{\xi^{(h_1+h_2)k} - 1} = 0$$

Similarly, $\sum_{k=0}^{n-1} \xi^{-(h_1+h_2)k} = 0$. Therefore,

$$\langle \chi_{h_1}, \chi_{h_2} \rangle_2 = \sum_{k=0}^{n-1} (\xi^{(h_1-h_2)k} + \xi^{(h_2-h_1)k})$$

If $h_1 \neq h_2$, then $0 < h_2 - h_1 < n$, so with similar argument, $\sum_{k=0}^{n-1} \xi^{(h_1-h_2)k} = \sum_{k=0}^{n-1} \xi^{(h_2-h_1)k} = 0$. Then

$$\langle \chi_{h_1}, \chi_{h_2} \rangle_2 = \begin{cases} |D_n| & h_1 = h_2 \\ 0 & h_1 \neq h_2 \end{cases}$$

Hence, by Corollary 6.26 and Theorem 6.25, these two-dimensional representations are irreducible and inequivalent. Following the decomposition of Corollary 6.26, and the fact that $4 \times 1^2 + (\frac{n}{2} - 1) \times 2^2 = 2n$, we have a complete list of inequivalent, irreducible representations of D_n .

Corollary 7.6. Let ψ_i, π_h be as in Prop. 7.5. (even n), let $\Gamma \subseteq D_n$. For each $i = 0, 1, 2, 3$, let

$$\lambda_{\psi_i} = \sum_{\gamma \in \Gamma} \psi_i(\gamma)$$

For each $1 \leq h \leq \frac{n}{2} - 1$, define

$$M_h = \sum_{\gamma \in \Gamma} \pi_h(\gamma)$$

Let $\lambda_h^{(1)}, \lambda_h^{(2)}$ be the eigenvalues of M_h . Then the spectrum of $\text{Cay}(D_n, \Gamma)$ is

$$\{\lambda_{\psi_1}, \dots, \lambda_{\psi_4}, \lambda_1^{(1)}, \lambda_1^{(2)}, \dots, \lambda_{n/2-1}^{(1)}, \lambda_{n/2-1}^{(2)}\}$$

Proof

This follows from Prop. 7.5. and Prop. 7.1. (i.e. eigenvalue as (eigenvalues from) sum of matrix representations over symmetric multiset).

7.5 Payley Graphs

Recall that a d -regular graph is Ramanujan if $\lambda(X) \leq 2\sqrt{d-1}$. Here $\lambda(X)$ denotes the largest eigenvalue of the graph. For an infinite sequence of Ramanujan graphs with fixed degrees, $h(X_n) \geq \frac{d-\lambda_1(X_n)}{2} \geq \frac{d-2\sqrt{d-1}}{2} > 0$ (first inequality from chapter 1)(last inequality if $d > 3$). So Ramanujan graphs are good candidates for expander family.

In this section, we construct an infinite sequence of Ramanujan graphs with varying degrees (so not exactly an expander family). One trivial case is K_n of complete graphs (whose spectrum is $n-1$ of multiplicity 1 and -1 of multiplicity $n-1$), this section discuss a non-trivial case.

Definition 7.7. For prime p , let

$$\mathbb{Z}_p^\times = \mathbb{Z}_p \setminus \{0\} \quad \Gamma_p = \{x^2 \mid x \in \mathbb{Z}_p^\times\}$$

Remark

- $\mathbb{Z}_p$ is a group defined with addition, while $\mathbb{Z}_p^\times$ is a group defined with multiplication. When we compute Γ , we use multiplication for x^2 , although later the Cayley graph is formed with $\mathbb{Z}_p$ where the operation is addition.
- x^2 corresponds to at most two elements of $\mathbb{Z}_p^\times$, namely x and $-x$. Suppose otherwise, $a^2 \equiv b^2 \pmod{p}$, then $(a-b)(a+b) \equiv 0 \pmod{p}$, thus either $a \equiv b \pmod{p}$ or $a \equiv -b \pmod{p}$ since p is prime. However, in general we don't know what these squares are exactly.

Lemma 7.8. If p is a prime and $p \equiv 1 \pmod{4}$, then there exists an $x \in \mathbb{Z}_p^\times$ s.t. $x^2 = -1$

Proof

- Since $\mathbb{Z}_p$ is a finite field, we have $\mathbb{Z}_p^\times$ is a cyclic group under multiplication (general result: for finite field $\mathbb{F}$, if H is a finite multiplicative subgroup of $\mathbb{F}^\times$, then H is cyclic).
- So $\mathbb{Z}_p^\times = \langle g \rangle$ for some $g \in \mathbb{Z}_p^\times$
- Consider $(x-1)(x+1) = 0$ in $\mathbb{Z}_p$. Clearly 0 is not the solution, so the solution lies in $\mathbb{Z}_p^\times$. Following discussions in the remark, only solutions are $x = 1, -1$. Since $g^0 = 1$ and $g^{(p-1)/2}$ both solve $x^2 = 1$, we have $g^{(p-1)/2} = -1$, so $g^{(p-1)/4}$ solves $x^2 = -1$.

Lemma 7.9. $|\Gamma_p| = \frac{p-1}{2}$

Proof

Consider the group homomorphism $\phi : \mathbb{Z}_p^\times \rightarrow \mathbb{Z}_p^\times$ by $\phi(x) = x^2$ which we use to construct Γ (homomorphism trivial). By previous Lemma, $\text{Ker}(\phi)$ contains 1, -1 only, so by first homomorphism theorem ($G/\text{Ker}(\phi) \cong \text{Im}(\phi)$), $|\text{Im}(\phi)| = |G/\text{Ker}(\phi)| = (p-1)/2$.

Proposition 7.10. Let p be a prime with $p \equiv 1 \pmod{4}$

- Γ_p is a symmetric subset of $\mathbb{Z}_p$
- The graph $\text{Cay}(\mathbb{Z}_p, \Gamma_p)$ is connected and $\frac{p-1}{2}$ -regular.

Remark

Graph $\text{Cay}(\mathbb{Z}_p, \Gamma_p)$ is called a Payley graph, a.k.a. *Paley*(p). There are infinitely many primes of the form $4k+1$ by Dirichlet's theorem, so Payley graphs indeed form an infinite family.

Proof

- By last lemma, there exists $x^2 = -1 \in \Gamma_p$, so the if $\gamma = y^2 \in \Gamma_p$, then $-\gamma = (xy)^2 \in \Gamma_p$
- Connected since $1 \in \Gamma_p$

Definition 7.11. Let $\xi = e^{2\pi i/p}$, then

$$G_p(a) = \sum_{k=0}^{p-1} \xi^{ak^2}$$

is called a Gauss sum.

Remark

By results of section 3, the eigenvalues of the Cayley graph $\text{Cay}(\mathbb{Z}_p, \Gamma_p)$ are given by

$$\lambda_a = \sum_{k \in \Gamma_p} \xi^{ak} = \frac{1}{2} \sum_{k=1}^{p-1} \xi^{ak^2} = \frac{1}{2} G_p(a) - \frac{1}{2}$$

where $a = 0, 1, \dots, p-1$ and $\xi = e^{2\pi i/p}$. So we focus on bounding $G_p(a)$ from above.

Definition 7.12. Let $a \in \mathbb{Z}$ and p be an odd prime. The Legendre symbol is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \text{ divides } a \\ 1 & \text{if } x^2 \equiv a \pmod{p} \text{ has a solution} \\ -1 & \text{if } x^2 \equiv a \pmod{p} \text{ has no solution} \end{cases}$$

Lemma 7.13. Let $a, b \in \mathbb{Z}$ and p an odd prime

- $\left(\frac{a^2}{p}\right) = 1$
- If $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$
- $\left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$

Proposition 7.14. If $a \not\equiv 0 \pmod{p}$, then

$$G_p(a) = \sum_{k=1}^{p-1} \left(\frac{k}{p}\right) \xi^{ak}$$

where $\xi = e^{2\pi i/p}$

Proof

Let $T_p = \mathbb{Z}_p^\times \setminus \Gamma_p$, which is the set of nonsquares, so $\left(\frac{k}{p}\right) = 1$ for all $k \in \Gamma_p$, $\left(\frac{k}{p}\right) = -1$ for all $k \in T_p$

$$\sum_{k=1}^{p-1} \left(\frac{k}{p}\right) \xi^{ak} = \sum_{k \in \Gamma_p} \xi^{ak} - \sum_{k \in T_p} \xi^{ak} = 2 \sum_{k \in \Gamma_p} \xi^{ak} + 1 - \sum_{k=0}^{p-1} \xi^{ak}$$

where 1 comes from taking $k = 0$. Take the sum of last term, it goes to 0, thus

$$\sum_{k=1}^{p-1} \left(\frac{k}{p}\right) \xi^{ak} = 2 \sum_{k \in \Gamma_p} \xi^{ak} + 1 = \sum_{k=1}^{p-1} \xi^{ak^2} = G_p(a)$$

Proposition 7.15. If $a \not\equiv (\text{mod } p)$, then

- $G_p(a) = \left(\frac{a}{p}\right) G_p(1)$
- $|G_p(a)| = \sqrt{p}$

Proof

- Since $\mathbb{Z}_p^\times$ is a multiplicative group, for all $a \in \mathbb{Z}_p^\times$, the map $a \rightarrow ak$ is a bijection (surjective: for t , take tk^{-1} ; injective: from mod). Hence

$$\left(\frac{a}{p}\right) G_p(a) = \left(\frac{a}{p}\right) \sum_{k=1}^{p-1} \left(\frac{k}{p}\right) \xi^{ak} = \sum_{k=1}^{p-1} \left(\frac{ak}{p}\right) \xi^{ak} = \sum_{t=1}^{p-1} \left(\frac{t}{p}\right) \xi^t = 1$$

second equality by Legendre symbol property, third equality by bijection

- We compute $G_p(1)^2$

$$\begin{aligned} G_p(1)^2 &= \left(\sum_{m=1}^{p-1} \left(\frac{m}{p}\right) \xi^m \right) \left(\sum_{n=1}^{p-1} \left(\frac{n}{p}\right) \xi^n \right) \\ &= \sum_{m=1}^{p-1} \sum_{n=1}^{p-1} \left(\frac{m}{p}\right) \xi^m \left(\frac{mn}{p}\right) \xi^{mn} \\ &\text{use the bijection } n \rightarrow mn \\ &= \sum_{m=1}^{p-1} \sum_{n=1}^{p-1} \left(\frac{m^2 n}{p}\right) \xi^{m+mn} \\ &= \sum_{m=1}^{p-1} \sum_{n=1}^{p-1} \left(\frac{n}{p}\right) \xi^{m+mn} \end{aligned}$$

break $m^2 n$, squared Legendre symbol gives 1

Take the sum, we have

$$\sum_{m=1}^{p-1} (\xi^{1+n})^m = \begin{cases} p-1 & n = p-1 \\ -1 & n = 1, 2, \dots, p-2 \end{cases}$$

Let T_p be the set of non-squares, we have $|\Gamma_p| = |T_p| = \frac{p-1}{2}$. Therefore

$$\begin{aligned} G_p(1)^2 &= - \sum_{n=1}^{p-2} \left(\frac{n}{p} \right) + (p-1) \left(\frac{p-1}{p} \right) \\ &= - \sum_{n=1}^{p-1} \left(\frac{n}{p} \right) + (p) \left(\frac{p-1}{p} \right) \\ &= - \left(\sum_{n \in \Gamma_p} 1 + \sum_{n \in T_p} (-1) \right) + p \left(\frac{p-1}{p} \right) \\ &= p \left(\frac{p-1}{p} \right) = \left(\frac{-1}{p} \right) p \\ &\text{under mod } p, -1 = p-1 \\ &= p \\ &\text{we know } x^2 = -1 \text{ has a solution} \end{aligned}$$

Hence, $|G_p(a)| = \sqrt{p}$.

Proposition 7.16. If $p \equiv 1 \pmod{4}$, then $\text{Cay}(\mathbb{Z}_p, \Gamma_p)$ is a Ramanujan graph.

Proof

Recall the graph is of degree $\frac{p-1}{2}$

$$|\lambda_a| = \left| \frac{G_p(a) - 1}{2} \right| \leq \frac{1}{2}(\sqrt{p} + 1) \leq 2\sqrt{\frac{p-1}{2} - 1}$$